

Release Summary

Product: CircleHome

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Key User Stories and Acceptance Criteria

(1.1) User Registration and Login - As a new user, I want to create an account with an email and password so that I can access my household securely.

Acceptance Criteria:

- User can sign up with email and password via the Sign Up screen
- User can sign in with existing credentials and is routed to the home screen
- Auth state persists across app restarts (users remain logged in)
- Form validation displays errors for invalid or missing input

(1.2) Household Creation and Join - As a new user, I want to create a household or join one using an invite code so that I can coordinate with the people I live with.

Acceptance Criteria:

- User can create a new household with a name; a unique invite code is generated and stored in Firestore
- User can join an existing household by entering a valid invite code
- Users can belong to multiple households and switch between them via the household switcher
- A newly joined user appears in the household member list

(1.3 / 2.1 / 2.2) Task Creation and Assignment - As a household member, I want to create a task with a title, category, difficulty, due date, and assigned member so that responsibilities are clearly defined.

Acceptance Criteria:

- Task creation form accepts title, category, due date/time, assignee, difficulty (1–5), and recurring options
- Assignee dropdown populates in real time from Firestore household members
- Auto-Assign selects the member with the lowest current workload score
- Tasks can be assigned to "Everyone" (creates individual tasks for each member)
- Task saves to Firestore with all correct fields

(1.4 / 2.3 / 2.7) Task Feed and Filtering - As a household member, I want to view and filter household tasks so that I can manage my responsibilities.

Acceptance Criteria:

- Task feed displays all household tasks, with a "My Tasks" toggle to show only tasks assigned to the current user
- Tasks can be filtered by status: incomplete, completed, overdue
- Overdue tasks are visually distinguished with color
- Empty states are handled for each filter

(1.5 / 2.4) Task Completion and Undo - As a household member, I want to mark a task complete and undo it if needed.

Acceptance Criteria:

- Tapping the complete button marks the task as completed and awards difficulty points to the assignee
- A snackbar with an Undo action appears after completion
- Tapping Undo reverses the completion and deducts the points

(3.2) Edit Task - As a household member, I want to edit a task so that I can update responsibilities without creating a new one.

Acceptance Criteria:

- Edit Task screen allows updating title, category, due date/time, assignee, difficulty, and recurrence settings
- Changes are saved to Firestore and reflected immediately in the task feed

(3.3) Notifications - As a household member, I want to receive notifications so that I stay aware of upcoming or overdue responsibilities.

Acceptance Criteria:

- App requests notification permission on first launch
- FCM token is saved to Firestore per user and refreshed on token rotation
- Foreground messages are displayed as in-app snackbars with title and body

(3.4) Dashboard and Recent Activity - As a household member, I want a cleaner and more informative dashboard so that household information is easier to view.

Acceptance Criteria:

- Home screen shows household name, member list, and a recent activity feed
- Recent activity displays the last 5 completed tasks with relative timestamps (e.g. "3h ago")
- Household creation/join flow is accessible from Settings, not the home screen
- Dashboard updates in real time via Firestore snapshot listeners

(3.5 / 4.3 / 4.5) Navigation and UI Consistency - As a household member, I want a consistent and easy-to-use interface so that navigating the app feels smooth and intuitive.

Acceptance Criteria:

- Bottom navigation bar is fixed across all main screens (Home, Tasks, Care, Leaderboard, Profile)
- Consistent pastel color theme and typography applied across all screens
- Loading and empty states are handled gracefully on all screens
- Sign-out confirmation dialog is present

(Care Diary) Care Notes - As a caregiver, I want to log care notes so that I can maintain a running health record for household members.

Acceptance Criteria:

- Add Note sheet accepts title, category (General, Medication, Appointment, Emergency), description, and an optional "About" field for the care recipient
- Notes appear in the shared Care Diary feed sorted by most recent
- Author name and care recipient are displayed on each entry
- Emergency notes are visually highlighted
- Notes can be deleted via a confirmation dialog

(4.1 / 4.4) Leaderboard and Weighted Scoring - As a household member, I want to see a task completion leaderboard so that I can track how much each person is contributing.

Acceptance Criteria:

- Leaderboard ranks all household members by total difficulty points earned from completed tasks
- A podium view (1st, 2nd, 3rd) is shown when 3 or more members exist, with avatars and medal icons
- Each ranked row shows member name, avatar, task count, and total points
- Rankings update in real time via Firestore stream
- Tapping a member navigates to their per-member contribution detail screen

(4.2) Per-Member Contribution Detail - As a household member, I want to view each member's task contributions so that fairness is transparent.

Acceptance Criteria:

- Contribution detail screen shows a breakdown of completed and pending tasks for the selected member
- Reachable by tapping any entry on the leaderboard

(Recurring Tasks) - As a user, I want to create a recurring task that repeats on a schedule.

Acceptance Criteria:

- Task creation and edit forms include a Recurring toggle and frequency selector (Daily, Every Other Day, Weekly, Biweekly, Monthly, Every 3 Months, Yearly)
- Recurring tasks display a recurring indicator in the feed
- After completion, a new instance is generated for the next recurrence date

(Task Swap) - As a user, I want to request a task swap with another household member.

Acceptance Criteria:

- From any assigned task, the user can open a swap sheet to request a swap or reassignment
- If the target accepts, the task is reassigned to them
- If rejected, the task remains with the original assignee

(Profile Setup) - As a user, I want to set my display name, profile photo, and workload level.

Acceptance Criteria:

- Profile screen allows setting display name, photo URL, and weekly workload level (1–5)
- Changes propagate across the app (task assignee labels, leaderboard avatars, care note authors)

Known Problems

Bugs:

- Push notifications for task events are not triggered from the app. Client-side FCM setup (permission request, token save, foreground snackbar) is implemented, but no Cloud Functions or server-side triggers exist to send notifications when a task is assigned, completed, or becomes overdue.

- Task completion fires immediately without a confirmation dialog. There is no confirmation prompt before marking a task complete — this was identified as a gap in Sprint 3 (3.2).
- Care notes are deletable. The Care Diary currently shows a delete button with a confirmation dialog. Editing is not allowed, but deletion is.
- Leaderboard score matching uses display name string instead of UID. If a member changes their display name, their completed tasks may not correctly attribute to them on the leaderboard.

Omissions (Missing Functionality):

- Background and terminated-state push notification handlers (onBackgroundMessage, onMessageOpenedApp) are not registered — notifications arriving when the app is backgrounded or closed are not processed.
- No scheduled overdue task reminder notifications (no Cloud Functions cron trigger).
- No image picker or Firebase Storage upload for profile photos — users must supply a photo URL manually.
- The full capacity-weighted Smart Rotation Engine (respecting each member's configured weekly capacity when auto-assigning) is not implemented. Auto-assign uses lowest task count only.
- SMS escalation for overdue care tasks is not implemented.
- AI-generated weekly care summary and PDF export are not implemented (planned as a future release feature).
- Task swap push notifications to the target member are not wired (same FCM gap above).

Design Shortcuts:

- Task categories (Cleaning, Groceries, Laundry, Bills, Other) and care categories (General, Medication, Appointment, Emergency) are hardcoded in the UI and cannot be customized per household.
- Graceful offline/slow network degradation was planned in Sprint 2 (2.5) but not fully implemented — the app shows loading spinners indefinitely when connectivity is lost.
- Firestore security rules for Sprint 3–4 collections (notificationTokens, swapRequests, careNotes) were flagged for audit in task 4.6 and may not be fully covered.

Product Backlog

The following high-priority user stories and bug fixes are recommended for a follow-on release:

1. Implement Firebase Cloud Functions to send push notifications on task assignment, task overdue, and swap requests — Bug Fix / Feature
2. Add background and terminated-state FCM handlers (onBackgroundMessage, onMessageOpenedApp) — Bug Fix
3. Switch leaderboard and task scoring to match members by UID instead of display name string — Bug Fix
4. Add task completion confirmation dialog before marking complete — Bug Fix
5. Add image picker + Firebase Storage upload flow for profile photos — Feature
6. Full capacity-weighted Smart Rotation Engine respecting each member's configured weekly capacity — Feature
7. Scheduled overdue task reminders via Cloud Functions (daily cron) — Feature
8. Offline mode: graceful UI degradation and queued writes when network is unavailable — Feature
9. Task swap push notifications to target member — Feature
10. Audit and update Firestore security rules for notificationTokens, swapRequests, and careNotes — Technical Debt
11. SMS escalation for overdue care tasks — Feature
12. AI-generated weekly care summary with PDF export for physician review — Feature
13. Customizable task and care categories per household — Feature
14. Full mutual-accept task swap flow (currently reassign-only; both-party confirmation not enforced) — Feature
15. Per-household notification preferences and profile aliases for multi-household users — Feature